Number 6 — Direct and inverse proportion · Fractional and negative indices

No calculator for questions 4 and 5. Marks in brackets. ★ = harder question.

1 The cost of copper wire is directly proportional to its length. A 2.5 m length costs £4.50. [4]

a Find the cost of 7 m of wire. **b** What length can be bought for £27?

c Write a formula connecting the cost C (in £) and the length L (in m).

> Find the cost of 1 m first: $4.50 \div 2.5 = £1.80$ per metre. **a** $7 \times 1.80 = £12.60$. **b** $27 \div 1.80 = 15$ m. **c** $C = 1.8L$.

2 The time taken for a journey is inversely proportional to the average speed. At an average speed of 60 km/h the journey takes 40 minutes. [4]

a How long does it take at 80 km/h?

b What average speed is needed to complete it in 32 minutes?

> Inverse proportion: the product speed × time is constant: $60 \times 40 = 2400$. **a** $2400 \div 80 = 30$ minutes. **b** $2400 \div 32 = 75$ km/h. (Check: faster speed, shorter time ✓.)

3 A tank can be emptied by 4 identical pumps in 9 hours. The time taken is inversely proportional to the number of pumps working. [4]

a How long would 6 pumps take?

b How many pumps are needed to empty the tank in 4.5 hours?

c Sketch the graph of time against number of pumps.

> Pumps × hours is constant: $4 \times 9 = 36$. **a** $36 \div 6 = 6$ hours. **b** $36 \div 4.5 = 8$ pumps. **c** A reciprocal curve: it falls steeply at first and flattens out, never touching either axis (time $= 36/n$).

4 Work out, without a calculator. Give exact answers. [9]

a $49^{\frac{1}{2}}$ **b** $27^{\frac{2}{3}}$ **c** $16^{-\frac{1}{2}}$ **d** $\left(\frac{25}{9}\right)^{\frac{3}{2}}$ **e** 5^{-2}

f $\left(\frac{2}{3}\right)^{-3}$ **g** $8^{-\frac{2}{3}}$ **h** $\left(1\frac{7}{9}\right)^{\frac{1}{2}}$ **i** $1 \div 81^{-\frac{1}{4}}$

Root first (the denominator of the index), then the power (the numerator); a negative index means "one over".

a $\sqrt{49} = 7$. **b** $(\sqrt[3]{27})^2 = 3^2 = 9$. **c** $\frac{1}{\sqrt{16}} = \frac{1}{4}$. **d** $\left(\sqrt{\frac{25}{9}}\right)^3 = \left(\frac{5}{3}\right)^3 = \frac{125}{27}$. **e** $\frac{1}{5^2} = \frac{1}{25}$.

f Flip the fraction, then cube: $\left(\frac{3}{2}\right)^3 = \frac{27}{8}$. **g** $\frac{1}{(\sqrt[3]{8})^2} = \frac{1}{4}$. **h** $1\frac{7}{9} = \frac{16}{9}$, so $\sqrt{\frac{16}{9}} = \frac{4}{3}$.

i $81^{-\frac{1}{4}} = \frac{1}{3}$, and $1 \div \frac{1}{3} = 3$.

5 Solve for x, without a calculator. [4]

a $\left(\frac{4}{9}\right)^x = \frac{27}{8}$ **b** $125^x = \frac{1}{25}$

Write both sides as powers of the same base, then equate the indices.

a $\frac{4}{9} = \left(\frac{2}{3}\right)^2$ and $\frac{27}{8} = \left(\frac{3}{2}\right)^3 = \left(\frac{2}{3}\right)^{-3}$. So $\left(\frac{2}{3}\right)^{2x} = \left(\frac{2}{3}\right)^{-3} \Rightarrow 2x = -3 \Rightarrow x = -\frac{3}{2}$.

b $125 = 5^3$ and $\frac{1}{25} = 5^{-2}$. So $5^{3x} = 5^{-2} \Rightarrow 3x = -2 \Rightarrow x = -\frac{2}{3}$.

Total: 25 marks

Check: direct proportion — find the value for 1 unit (or keep the ratio constant) · inverse proportion — the product is constant · $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$: root first · $a^{-n} = \frac{1}{a^n}$ and $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$ · to solve $a^{f(x)} = a^{g(x)}$, equate the indices.